

Introduction to VRP

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1 Defining the VRP

In the previous section I give a motivation for why to be interested in Volatility as an asset class and more specifically be interested in this concept of a volatility risk premium. Below, I define the volatility risk premium as being the difference between the implied volatility over the next t days, minus the realised volatility over the next t days. This makes intuitive sense as we have discussed that investors are willing to pay extra for downside protection.

$$VRP = IV - RV$$

Here we have our definition for the volatility risk premium. We have two terms, firstly, the implied volatility followed by the realised volatility. Let us, start by defining the implied volatility.

$$IV = \arg \min_{\sigma > 0} |C_{BS} - C_{mrkt}|$$

You are choosing a value through, often Newton-Raphson, numerical methods to minimise the difference between your model price and the market price. This makes sense as the you are using market prices and the information about future volatility, to price other options. We define our realised volatility as being

$$RV = \sqrt{\sum_{t=1}^T \ln\left(\frac{S_t}{S_{t-1}}\right)^2}$$

Now, we want to be able to trade this concept, we are looking for a contract with a payoff based off the difference between the implied volatility and the realised. Variance swaps are an obvious choice for this. In the next part I introduce variance swaps and justify my use for the VIX as a 'fair strike' for the swap.